SIMATS ENGINEERING

ASSESSMENT TOOL II – REVERSE ENGINEERING TASK (MEDIUM)

Course Name: Ethical Hacking for Secure Networks	**Code:** ITA1436	**CO:** CO2
Duration: 90 min **Date:**	**Total Marks:**	**Reg. No / Name :** Dinesh R-192421117

CO2: *Execute footprinting in social engineering such as DNS analysis, competitive intelligence gathering, and attack simulations to collect relevant information for identifying potential security risks.(BL3)*
Assessment Tool Weightage: 30%

Instructions: *Answer all questions. Each question carries 10 marks.*

Task: 1 A cybersecurity team discovers a suspicious executable named **FootPrintX.exe** on a penetration tester's workstation. The application appears to collect information about target organizations, including domain names, IP addresses, and network details. The source code is unavailable, and the analyst must reverse engineer the binary to understand its functionality. Your task is to identify how the tool performs footprinting activities.

Questions:

1. Which functions or modules are responsible for gathering target information?
Functions such as DNS lookup, IP address scanning, and network information collection modules gather target information.

2. What footprinting techniques are implemented in the executable?
The tool may use DNS enumeration, IP address lookup, WHOIS queries, and network scanning to collect information about the target.

3. How can the information collection process be disabled or modified?
The collection process can be disabled by removing or disabling the relevant functions, blocking unauthorized network requests, and updating security rules.

Task: 2 A file named **DNSScan.exe** has been found during a security audit. Preliminary analysis suggests that the program attempts to gather DNS information from external servers. Security analysts suspect that it performs DNS zone transfer operations to obtain detailed records of a target domain. Reverse engineer the executable to determine its behavior and associated risks.

Questions:

1. Which function is responsible for initiating DNS requests?
A DNS query function, such as getaddrinfo() or a DNS resolver module, may initiate DNS requests.

2. Does the application attempt an AXFR (Zone Transfer) request? Justify your answer.
The application attempts an AXFR request if its code contains DNS query logic specifying the AXFR record type. This must be confirmed through reverse engineering or network traffic analysis.

3. What information can an attacker obtain using the extracted DNS records?
An attacker may discover subdomains, IP addresses, mail servers, name servers, and internal hostnames exposed through DNS records.

Task: 3 An organization identifies a tool called **IntelGather.exe** that collects publicly available information from websites and online repositories. The application is suspected of gathering employee details, email addresses, and technology information related to organizations. Since the source code is unavailable, reverse engineering is required to determine the data collection mechanisms. Analyze the executable and identify its intelligence-gathering capabilities.

Questions:

1. What types of information are collected by the application?
The application may collect employee names, email addresses, job roles, company details, technologies used, and publicly available organizational information.

2. Which external websites or services are accessed during execution?
It may access search engines, company websites, professional networking sites, public code repositories, and WHOIS services. The exact services must be identified through network analysis.

3. How can the collected intelligence be misused by attackers?
Attackers may use the information for targeted phishing, impersonation, social engineering, password attacks, and identifying potential vulnerabilities.

Task: 4 A suspicious program named **MailCraft.exe** is discovered on a compromised system. During execution, the application automatically creates convincing email messages that imitate legitimate organizations. Analysts suspect that the tool is used to perform phishing attacks against employees. Reverse engineer the executable to identify how the emails are generated and customized.

Questions:

1. Which functions are responsible for generating phishing emails?
Email template generation, message customization, and email-sending functions may be responsible for creating phishing emails.

2. What information is used to personalize the messages?
The application may use employee names, email addresses, job titles, company names, and information about the impersonated organization.

3. How can the phishing functionality be prevented or removed?
Remove the malicious application, block unauthorized email-sending activity, update endpoint protection, and configure email filtering and authentication mechanisms such as SPF, DKIM, and DMARC.

Task: 5 A binary file named **PortalClone.exe** imitates the login page of a popular online service. Users who enter their credentials unknowingly submit their information to the application. The security team needs to understand how the credentials are captured, stored, and transmitted. Reverse engineer the program to identify its malicious behavior.

Questions:

1. How does the application capture usernames and passwords?
The application may use a fake login page containing input fields that collect usernames and passwords when users submit the form.

2. Where are the captured credentials stored or transmitted?
The credentials may be stored in local files or databases, or transmitted to a remote server controlled by the attacker. This can be verified by examining the executable and monitoring network traffic.

3. What security risks arise from the use of such applications?
These applications can cause credential theft, account hijacking, identity theft, financial loss, and unauthorized access to sensitive information.

Code of Conduct:

I R Dinesh-192421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dinesh R

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of System	20%				
Decomposition	25%				
Identification of Components	20%				
Reconstruction	20%				
Documentation	15%				

Criteria	Understanding of System	Decomposition	Identification of Components	Reconstruction	Documentation	Total
Q1						
Q2						
Q3						
Q4						
Q5						